

Executive Business Report

Sales Forecasting & Demand Intelligence System

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

Executive Summary

This report presents an analysis of the company's historical sales to support better inventory planning and business decision-making. The system identifies future sales trends, highlights unusual sales patterns, and groups products according to their demand behaviour. These insights help reduce excess inventory, prevent stock shortages, and improve purchasing decisions. Overall, the results provide a practical way to improve operational efficiency and increase profitability.

Key Findings

- Sales show clear fluctuations throughout the year, indicating seasonal demand.
- Some product categories consistently generate much higher sales than others.
- A few months experienced unusually high or low sales, which may require further business investigation.
- Products can be divided into four demand groups, allowing different stocking strategies for different products.
- The sales forecast suggests relatively stable demand over the next three months.

Three-Month Sales Forecast

Month	Expected Sales	Business Interpretation
Month 1	82843.34375	Sales expected to remain stable.
Month 2	80870.921875	Similar demand is expected.
Month 3	85361.2265625	Continue current inventory planning.

The forecast provides an estimate rather than an exact number. Actual sales may vary depending on promotions, holidays, or market conditions.

Major Sales Anomalies

Three unusual sales periods were identified:

Period	Observation	Possible Reason
February 2015	Very low sales	Lower customer demand or stock shortage
February 2016	Significant drop	Supply or operational issue
September 2018	Very high sales	Festival season, promotion, or bulk orders

These events should be reviewed to understand their causes and improve future planning.

Product Demand Segmentation

Products were grouped into four demand categories:

- High Volume, Stable Demand: Maintain sufficient inventory because these products sell consistently.
- Growing Demand: Increase stock gradually as customer demand is rising.
- Low Volume, High Volatility: Keep limited inventory to reduce storage costs.
- Declining Demand: Reduce future purchasing and consider promotional offers to clear existing stock.

Business Recommendations

1. Use sales forecasts to plan purchases so inventory is available before demand increases.
2. Maintain higher stock for products with consistent demand to avoid lost sales caused by stock shortages.
3. Review unusual sales changes regularly to quickly identify supply problems, seasonal trends, or successful promotions.

Limitation

The forecast is based on historical sales patterns. Unexpected events such as market changes, supplier delays, or special promotional campaigns may affect actual sales. Therefore, business decisions should combine these insights with managerial experience.

Note: I used AI assistance to develop parts of the Streamlit interface and machine learning code while learning these technologies. I reviewed, understood, and adapted the generated code throughout the project, making it a valuable learning experience.